

Introduction to Artificial Intelligence

Artificial Intelligence (AI) is one of the fastest-growing fields in computer science.

It focuses on creating intelligent systems that can perform tasks requiring human intelligence, such as problem-solving, learning, reasoning, and decision-making.

Machine Learning is a major branch of AI that enables systems to learn from data.

Instead of being explicitly programmed, machine learning models improve their performance through experience and training data.

There are three main types of machine learning:

1. Supervised Learning
2. Unsupervised Learning
3. Reinforcement Learning

Deep Learning is a specialized area of machine learning that uses artificial neural networks with multiple hidden layers. Deep learning is widely used in image recognition, speech processing, and natural language understanding.

Natural Language Processing (NLP) helps computers understand and generate human language.

Applications of NLP include chatbots, virtual assistants, text summarization, sentiment analysis, and language translation.

Generative AI models such as GPT and BERT have transformed the AI industry. These models can generate

human-like text, answer questions, create summaries, and assist developers in writing code.

MLOps combines Machine Learning and DevOps practices to automate model training, deployment, monitoring, and scaling. It plays a critical role in deploying AI applications into production environments.

As AI technology continues to evolve, it is creating new opportunities across industries including healthcare, finance, cybersecurity, education, and autonomous systems.